Supplementary Information for

How aggregation can conceal composition: aggregate biocapacity and the identifiability of modeled ecological-capital drawdown

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October 24, 2026

This document supplements the main text (referred to as "the paper"). Section S1 gives the full symbol table and extension status; S2 the derivations and proofs behind Propositions 1–3 and the macro-ratio identities; S3 the numerical protocol and verification record; S4 extended model results (yield sweep, drawdown-depth ladder, delay and restoration checks, recovery metric and gate-sign phase dependence, regime table); S5 the empirical details (data vintages, Divisia and level decompositions, ϕ_0-robustness, cropland-book calibration, observation-operator hierarchy, regeneration anchors, recursive identifiability, and a physical-stock scoreboard); S6 the policy extensions; S7 the one-stock comparator and the quantities that belong to it rather than to the two-land model; S8 the controlled technology-wave/no-wave comparison. Equation numbers without the prefix S refer to the paper.

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S1 Symbol table and stated extensions

Following GFN/NFA practice, each annual footprint or biocapacity account value is reported in gha, not gha yr^{-1} (Global Footprint Network, 2009, 2019). With continuous model time measured in years, B, E, Y_f, Y_c, S have units gha; their time derivatives have units gha yr^{-1}. The leaky debt state D is the time-integral of the shortfall and has units gha yr, so $\dot{D}$ and ηD are gha. This internal model liability is not an NFA account variable.

Table S1: Symbols, meanings, units and roles. Per-hectare annual account yields are in gha ha^{-1}; $b_{G,c}$ is a regeneration-value coefficient in gha $\mathrm{yr}\,\mathrm{ha}^{-1}$; annual biocapacity, footprint and shortfall account amounts are in gha; the leaky debt state is in gha yr.

Symbol	Meaning	Unit	Role
A_f	fast provisioning land (cropland, intensive)	ha	state
A_c	ecological-capital land (forest, soil, wetland, long-rotation systems; effective class)	ha	state
A_r	reserve land, $A_{tot} - A_f - A_c$	ha	dependent balance
q	per-hectare capital-land quality (capacity index)	dimensionless	state
P	human population	cap	state
D	leaky accumulated ecological debt (time-integral of shortfall)	gha yr	state
b_f	fast-land account yield per hectare, $(b_{f0} + T_b)e^{-\alpha D}$	gha ha^{-1}	derived
$\tilde{b}_f(t)$	FAOSTAT physical crop output per hectare (proxy; normalized to obtain b_f)	product ha^{-1} yr^{-1}	empirical proxy
b_{f0}	base fast-land yield per hectare	gha ha^{-1}	parameter
b_c	capital-land direct yield	gha ha^{-1}	parameter
$b_{G,c}$	capital regeneration-value coefficient	gha $\mathrm{yr}\,\mathrm{ha}^{-1}$	parameter
$G_c(q)$	per-ha regeneration of quality, $\rho_c q(1 - q/q_{\max})$	yr^{-1}	rate function
$\Upsilon_c(q)$	per-ha capital value, $b_c + b_{G,c}G_c(q)$ (also written $b_{c,\mathrm{eff}}$)	gha ha^{-1}	derived
Δb_{conv}	composition diagnostic, $b_f - \Upsilon_c(q)$	gha ha^{-1}	derived
Y_f, Y_c	fast and capital yields, $b_f A_f$ and $\Upsilon_c A_c$	gha	derived
B	biocapacity, $Y_f + Y_c$	gha	aggregate
E	footprint account value, eP (or frozen)	gha	aggregate

Symbol	Meaning	Unit	Role
$X,\ C$	fast-book proxy account and accounting residual, $B = X + C$	gha	empirical decomposition
S	shortfall, $[E - \sigma_f Y_f - \sigma_c Y_c]_+$	gha	derived
u_c	conversion rate, $\min(S/b_f,\ A_c - A_c^{\min})/\tau_{\text{conv}}$	$\text{ha}\,\text{yr}^{-1}$	flow
K	carrying capacity, $\max(B/e, K_{\min})$	cap	algebraic
e	per-capita footprint	$\text{gha}\,\text{cap}^{-1}$	parameter
r	intrinsic population growth rate	yr^{-1}	parameter
ρ_c	quality regeneration rate	yr^{-1}	parameter
$q_{\max}$	quality ceiling	dimensionless	parameter
μ	cropland maintenance rate (reserve→fast)	yr^{-1}	parameter
η_f	fast-land retirement rate (fast→reserve)	yr^{-1}	parameter
τ_{conv}	conversion time constant	yr	parameter
α	degradation rate of b_f per unit debt	$\text{gha}^{-1}\,\text{yr}^{-1}$	parameter
η	debt-repayment rate	yr^{-1}	parameter
$\tau_g,\ \tau_p$	regeneration and demographic delays	yr	parameter
$\sigma_f,\ \sigma_c$	human-available shares of the two flows	dimensionless	parameter
R_{rc}	surplus-gated restoration, $\chi_r A_r\,[B - E]_+$ (reserve→capital)	$\text{ha}\,\text{yr}^{-1}$	gated flow
χ_r	restoration coefficient	$\text{gha}^{-1}\,\text{yr}^{-1}$	parameter
T_b	technology yield increment, $\omega_{Tb}\Delta b/(1 + e^{-\kappa_w(t - t_{wave})})$	$\text{gha}\,\text{ha}^{-1}$	off except in SI S8
ω_{Tb}	activation flag for the technology wave	dimensionless	0 baseline; 1 in SI S8
$\Delta b,\ \kappa_w,$ t_{wave}	amplitude, steepness and center of the wave	$\text{gha}\,\text{ha}^{-1}$; yr^{-1}; yr	active only in SI S8
$A_c^{\min},\ A_r^{\min},$ $A_f^{\max},\ K_{\min}$	capital viability floor, reserve floor, arable ceiling, carrying-capacity floor	ha, ha, ha, cap	typed floors
A_{tot}	total bioproductive area	ha	parameter
k	$\mu/(\mu + \eta_f)$, fast share of non-capital land at equilibrium	dimensionless	derived
$B_{eq}(A_c)$	frozen-demand balance branch, $b_{f0} k A_{tot} - (b_{f0} k - b_c) A_c$	gha	derived
E_{ceil}	demand ceiling, $B_{eq}(A_c^{\min})$	gha	derived
$\lambda_{1,2}$	deficit-side Jacobian eigenvalues	yr^{-1}	derived
R_B, R_A, ψ_f	E/B, E/Y_f, Y_f/B (so $R_A = R_B/\psi_f$)	dimensionless	diagnostics
ϕ_0, w_X, w_C	base-year cropland share and log-mean weights	dimensionless	empirical decomposition
$s_X, \bar{s}_X, \bar{s}_C$	current and average fast-book shares	dimensionless	empirical decomposition
$\mathcal{R}_c(T)$	recovery metric, $(A_c(T) - A_c^{\text{deg}})/(A_c^{\text{init}} - A_c^{\text{deg}})$	dimensionless	derived
$R_{fc},\ \chi$	fast→capital restoration variant, $\chi A_f\,[B - E]_+$	$\text{ha}\,\text{yr}^{-1}$; $\text{gha}^{-1}\,\text{yr}^{-1}$	extension (off)
$L_c,\ H_c$	timber liquidation flow $[H_c - \sigma_c Y_c]_+ / b_{G,c}$ and its target	$\text{ha}\,\text{yr}^{-1}$; gha	extension (off)

Extension status. (i) A timber-liquidation channel $L_c = [H_c - \sigma_c Y_c]_+ / b_{G,c}$ removes capital area when an annual timber-account target H_c exceeds the harvestable capital account amount; $\gamma = 1/b_{G,c}$ (ha gha^{-1} yr^{-1}) applies only to this channel and remains off. (ii) A direct fast-to-capital restoration $R_{fc} = \chi A_f [B - E]_+$ (surplus-gated) can replace or supplement R_{rc} and remains off. (iii) A reservation $\sigma_f, \sigma_c < 1$ withholds a share of each flow from demand (Section S6) and remains off. (iv) The technology wave T_b of Eq. (14) is off in the baseline, main figures and tables, and recovery-gate experiment; it is activated for the controlled matched comparison in SI §S8. The separate wave-on recovery case mentioned in S4.4 is a solver-regression check only and is excluded from the reported scenario set. The v44 solver propagates the wave consistently through current B, the returned account series, and delayed carrying-capacity history. (v) An endogenous per-capita footprint $e(t)$ is an offered extension not implemented here.

Conventions. The positive part is exact; conversion is a rate through τ_{conv}; land is conserved one-to-one across A_f, A_c, A_r while q is a separate capacity book; annual NFA account amounts B, E, Y_f, Y_c, S are in gha, while the leaky time-integral D is in gha yr and $\dot{D}, \eta D$ are in gha; a sustainable fast book exists because of the maintenance term μA_r. The reserve floor $A_r^{\min}$ and arable ceiling $A_f^{\max}$ are declared domain constraints that are checked after each run (never violated in any reported run: the minimum reserve in the drawdown sweeps is 0.80 ha) rather than enforced by clamps; $A_c^{\min}$, $A_f \geq 0$, $q \in [0, q_{\max}]$ and $K_{\min}$ are enforced.

S2 Derivations and proofs

S2.1 The composition identity with varying debt and quality

Differentiating $B = b_f A_f + \Upsilon_c(q) A_c$ with $b_f = (b_{f0} + T_b) e^{-\alpha D}$ and substituting Eqs. (6)–(7) gives the general identity

$$\dot{B} = \Delta b_{\text{conv}}\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c} G_c'(q) A_c\, \dot{q} + \Upsilon_c R_{rc} - \alpha \dot{D}\, b_f A_f + \frac{\dot{T}_b}{b_{f0} + T_b}\, b_f A_f. \quad \text{(S1)}$$

The conversion term has the sign of Δb_{conv}; the quality term has the sign of $G_c'(q)$ (non-negative for $q \leq q_{\max}/2$, non-positive above it, since $\dot{q} \geq 0$) and is bounded in magnitude by $b_{G,c} \rho_c \dot{q} A_c$, which is small at the baseline; the debt term is non-positive while debt accrues; the technology term is non-negative while the wave rises; the restoration term is non-negative. Proposition 1 of the paper is the case $T_b \equiv 0$, $R_{rc} \equiv 0$, D fixed. In the demonstration run D is not fixed (it rises to 0.20 gha yr and decays), so the debt term is present, and the mask holds nevertheless: the conversion term dominates by two orders of magnitude at the start ($\Delta b_{\text{conv}} u_c \approx 0.78$ gha ha^{-1} $\times$ 0.27 ha yr^{-1} = 0.21 gha yr^{-1} against $b_f(\mu A_r - \eta_f A_f) = -0.005$ gha yr^{-1} and $\alpha \dot{D} b_f A_f \approx 0.004$ gha yr^{-1}).

S2.2 The deficit-side affine system and its spectrum

On the deficit side with $q = q_{\max}$, D frozen, $R_{rc} = 0$ and $S/b_f < A_c - A_c^{\min}$, Eq. (5) gives $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\text{conv}})$, and with $A_r = A_{tot} - A_f - A_c$ Eqs. (6)–(7) become

$$\frac{d}{dt}\begin{pmatrix} A_f \\ A_c \end{pmatrix} = \begin{pmatrix} -\tau_{\text{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\text{conv}}} \\ \tau_{\text{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\text{conv}}} \end{pmatrix} \begin{pmatrix} A_f \\ A_c \end{pmatrix} + \begin{pmatrix} \dfrac{E}{b_f \tau_{\text{conv}}} + \mu A_{tot} \\ -\dfrac{E}{b_f \tau_{\text{conv}}} \end{pmatrix}. \quad \text{(S2)}$$

Hence $\operatorname{tr} J = -\tau_{\text{conv}}^{-1}(1 - b_c/b_f) - \mu - \eta_f$ and $\det J = (\mu + \eta_f)(k - b_c/b_f)/\tau_{\text{conv}}$ with $k = \mu/(\mu + \eta_f)$. At the baseline $\operatorname{tr} J = -1.0512\,\text{yr}^{-1}$ and $\det J = 0.05353\,\text{yr}^{-2}$, and the eigenvalues are

$$\lambda_{1,2} = \tfrac{1}{2}\Big(\operatorname{tr} J \pm \sqrt{\operatorname{tr}^2 J - 4 \det J}\Big) = -0.0537\,\text{yr}^{-1},\ -0.9975\,\text{yr}^{-1}. \quad \text{(S3)}$$

Because J contains neither E nor the state, the spectrum is constant along the frozen-demand branch (verified at $E/E_{\text{ceil}} = 0.50, 0.90, 0.99, 0.999$: $\lambda_1 = -0.05366\,\text{yr}^{-1}$ in every case). J contains no ρ_c, so λ_1 is exactly independent of the regeneration rate ($-0.0537\,\text{yr}^{-1}$ at $\rho_c = 0.04, 0.08, 0.16\,\text{yr}^{-1}$). It depends mildly on τ_{conv} (-0.0553, -0.0537, -0.0505, $-0.0417\,\text{yr}^{-1}$ at $\tau_{\text{conv}} = 0.5, 1, 2, 5\,\text{yr}$) and more strongly on the maintenance rate (-0.0254, -0.0537, $-0.1099\,\text{yr}^{-1}$ at $\mu = 0.03, 0.06, 0.12\,\text{yr}^{-1}$); its value should therefore be reported as an order-of-magnitude negative number that never crosses zero, while the qualitative statements (no positive eigenvalue, no fold, no critical slowing) hold across all sweeps. The fixed point of the affine system is $A_c^*(E)$ of Eq. (19) with $A_f^* = k(A_{tot} - A_c^*)$; for $E > E_{\text{ceil}}$ it lies below $A_c^{\min}$ and the trajectory approaches the face $A_c = A_c^{\min}$ asymptotically, with limiting $u_c = 0$ and persistent shortfall $S^* = E - E_{\text{ceil}}$. Two verified members of the no-conversion family at the baseline are $(A_f^*, A_c^*, A_r^*, P^*) = (0.914\,\text{ha}, 0.824\,\text{ha}, 0.762\,\text{ha}, 1.488\,\text{cap})$ and $(1.091\,\text{ha}, 0.500\,\text{ha}, 0.909\,\text{ha}, 1.731\,\text{cap})$, both with $A_r^*/A_f^* = \eta_f/\mu = 0.8333$ exactly; the second is a sustainable cropland equilibrium supported by the maintenance term. The slow mode is the equilibration of the reserve share $A_r/A_f \to \eta_f/\mu$, which is why the strict-inequality window of the demonstration run extends to $\approx 150\,\text{yr}$ although most of the conversion is complete within $25\,\text{yr}$.

In the other branch of the minimum ($S/b_f \geq A_c - A_c^{\min}$), $u_c = (A_c - A_c^{\min})/\tau_{\text{conv}}$ and the distance to the floor obeys $\frac{d}{dt}(A_c - A_c^{\min}) = -(A_c - A_c^{\min})/\tau_{\text{conv}}$: it decays exponentially and never reaches zero in finite time. This is the reason the floor is approached asymptotically in Fig. 1c of the paper and the operational floor-proximity thresholds depend on the tolerance used to declare proximity (Section S4.2).

S2.3 Characteristic structure of the full delay system

With state $z = (A_f, A_c, q, P, D)$ the linearized system about any point of a given active set is $\dot{z}(t) = J_0 z(t) + J_g z(t - \tau_g) + J_p z(t - \tau_p)$ with characteristic equation $\det \Delta(\lambda) = 0$, $\Delta(\lambda) = \lambda I - J_0 - J_g e^{-\lambda\tau_g} - J_p e^{-\lambda\tau_p}$. J_0 carries the conversion and debt feedbacks on the current state (the 2×2 block above, plus the D row $\partial\dot{D}/\partial z = (-b_f, -\Upsilon_c, -b_{G,c} G_c' A_c, e, -\eta)$ on the deficit side and the $\partial b_f/\partial D = -\alpha b_f$ couplings); J_g has the single entry $G_c'(q^*)$ in the q row; J_p has the row $(rP^{*2}/K^{*2})\, e^{-1}\, \partial B/\partial z$ in the P position (from $\partial\dot{P}/\partial K = rP^2/K^2$ and $K = B/e$). Two scalar delayed modes decouple at the baseline. The quality mode linearized at $q_{\max}$ is $\dot{\xi} = -\rho_c \xi(t - \tau_g)$ (rate $-0.08\,\text{yr}^{-1}$ without the delay); its characteristic root satisfies $\lambda = -\rho_c e^{-\lambda\tau_g}$, and since $\rho_c \tau_g = 1.6 > \pi/2$ the linear delayed mode would be oscillatory. In the nonlinear model this cannot occur because $\dot{q} = G_c(q(t - \tau_g)) \geq 0$ for $q(t - \tau_g) \in [0, q_{\max}]$ and q is clamped at $q_{\max}$, so quality is monotone non-decreasing and converges to $q_{\max}$; the linear oscillation is an artifact of linearizing a one-sided law at its boundary. The population mode on the no-conversion face is the delayed logistic law with $r\tau_p = 0.5 < \pi/2$, below the Hutchinson threshold, so it is non-oscillatory at the baseline. No entry, determinant sign or eigenvalue is inherited from the one-stock characteristic equation of Section S7; each is derived per active set (no conversion; active conversion; floor face $A_c = A_c^{\min}$; $K = K_{\min}$).

S2.4 Debt fixed points beyond the ceiling

For $E > E_{\text{ceil}}$ the limiting capital state is $A_c = A_c^{\min}$, approached asymptotically, and the persistent shortfall feeds the debt. With $B_{eq}(A_c^{\min}; D) = b_{f0} e^{-\alpha D} k(A_{tot} - A_c^{\min}) + b_c A_c^{\min}$, the debt equation $\dot{D} = E - B_{eq}(A_c^{\min}; D) - \eta D$ has the fixed point $D^*(E)$ solving $\eta D = E - B_{eq}(A_c^{\min}; D)$; the right-hand side is increasing in D, so the fixed point is unique and attracting whenever $\eta > \alpha b_{f0} k(A_{tot} - A_c^{\min})$ ($0.034\,\text{yr}^{-1} < 0.05\,\text{yr}^{-1}$ at the baseline). Solving numerically, $D^* = 1.186\,\text{gha yr}$ at $E = 1.158\,\text{gha}$ and $D^* = 5.375\,\text{gha yr}$ at $E = 1.238\,\text{gha}$, with $B(D^*) = 1.099$ and $0.969\,\text{gha}$ and persistent shortfalls $S^* = 0.059$ and $0.269\,\text{gha}$; the integrated debt values quoted in earlier versions (1.21 and $5.39\,\text{gha yr}$) agree to within 2% (finite horizon). Under frozen demand

the debt therefore saturates; it is unbounded only if the shortfall is.

S2.5 Boundedness asymmetry and the crossover shortfall

From Eq. (12), $\ln b_f = \ln(b_{f0} + T_b) - \alpha D$. The first term lies in $[\ln b_{f0},\ \ln(b_{f0} + \Delta b)]$, so the technology channel contributes at most $\ln(1 + \Delta b/b_{f0}) = \ln(1.85/0.85) = 0.778$ to $\ln b_f$ at the baseline. From Eq. (13), $\dot{D} \le \bar{S} - \eta D$ with $\bar{S}$ the running supremum of the shortfall, so $D(t) \le \max\{D(0), \bar{S}/\eta\}$ and the degradation channel contributes at least $-\alpha\bar{S}/\eta$. The two are equal when $\bar{S} = S^\dagger = (\eta/\alpha)\ln(1 + \Delta b/b_{f0}) = 1.296\,\text{gha}$ at the baseline. For the fast-book yield itself, $Y_f = b_f A_f$ shares this structure; the capital term $\Upsilon_c A_c$ is unaffected by either channel. Under the additive law $b_f = b_{f0} + T_b - \alpha D$, both channels are bounded quantities of the same kind (Δb against αD^*) and the logarithmic asymmetry disappears, which is why the paper states the result for the multiplicative form.

S2.6 Macro-ratio identities

Define $R_B = E/B$, $R_A = E/Y_f$ and the fast-land share $\psi_f = Y_f/B$; then $R_A = R_B/\psi_f$ is an identity. The direct-yield share inside the capital book, $\psi_c = b_c A_c/Y_c$, is a different quantity and $R_A = R_B/\psi_c$ is false. $R_B = 1$ is aggregate balance. With baseline shares $\sigma_f = \sigma_c = 1$, $R_A > 1$ (demand exceeds the fast-book account yield) is necessary but not sufficient for conversion, because the shortfall requires $E > B = Y_f + Y_c$, not merely $E > Y_f$; with reserved flows the threshold changes. The capital book declines iff $u_c > R_{rc}$, with u_c given by Eq. (5). Under the baseline shares, active conversion therefore requires $R_B > 1$; in the stated reservation extension $\sigma_f, \sigma_c < 1$, conversion and a declining capital book can occur while $R_B \le 1$, so aggregate overshoot is not required in that extension. "$R_B = 1$ is the fold" belongs to the one-stock frozen-demand reducer (Section S7), not to the two-land model. At the regeneration maximum $q^* = q_{\max}/2$ the direct-yield share of capital value is $b_c/(b_c + b_{G,c}\rho_c q_{\max}/4) = 0.758$ at the baseline; this is a property of that point, not a structural constant.

S3 Numerical protocol and verification record

Solver. Method of steps with an explicit first-order (Euler) update; exact positive part (a smooth ramp leaks conversion when $E < B$); clamps $A_c \ge A_c^{\min}$, $A_f \ge 0$, $q \in [0, q_{\max}]$, $P \ge 0$, $D \ge 0$, $K \ge K_{\min}$; the reserve is computed as $A_r = A_{tot} - A_f - A_c$ after each update and the conservation residual $|A_f + A_c + A_r - A_{tot}|$ is recorded at every step (maximum 2.2×10^{-16} ha in the baseline runs). The pre-run histories of B (for $K(t-\tau_p)$) and q (for $G_c(q(t-\tau_g))$) are constant at the initial state over the full interval $[-\max(\tau_g, \tau_p), 0)$; $B(0)$ is calculated from the model, never inserted as zero. The v44 solver also propagates the active logistic T_b into current B, returned output and delayed B history. In the earlier solver, a zero B sample at $t = 0$ caused an artificial dip in delayed carrying capacity, and the active wave was omitted from returned and delayed B; both bookkeeping errors are corrected. Step sizes: $\Delta t = 0.05\,\text{yr}$ for the demonstration run and the bisection thresholds; $\Delta t = 0.1\,\text{yr}$ for the drawdown-depth sweeps (checked against 0.05 at selected points; the operational initial-population floor-proximity threshold at $A_{c0} = 1.0\,\text{ha}$ moves from 2.955 to 2.98 cap, which is why thresholds are reported as bands).

Delayed versus current quality in the regeneration term. The deposited solver (`work/twoland_fixed.py`) computes the delayed quality $q(t - \tau_g)$ but evaluates the update of q at the current quality. Table S2 compares the two variants at the baseline; because q is monotone and the regenerative term is small ($b_{G,c}G_c \le 0.016\,\text{gha ha}^{-1}$ against $b_c = 0.05\,\text{gha ha}^{-1}$), the differences are below 1% in every reported quantity. The corrected variant (`work/twoland_fixed_delayed.py`, a one-line change) is deposited alongside; all quantities in the paper are quoted to a precision at which the two agree.

Table S2: Effect of evaluating the regeneration term at the delayed rather than the current quality (baseline registry). Biocapacity B, fast-book contribution Y_f, and capital contribution $\Upsilon_c(q)A_c$ are in gha; land areas A_c, A_f, A_{c0} are in ha; the operational initial-population floor-proximity threshold P_0 is in cap; delays and step intervals are in yr.

Quantity	current q (deposited)	delayed $q(t-\tau_g)$
Demonstration window with $\dot{B}>0$, $\dot{A}_c<0$ (yr)	150.3	149.0
Mask-end values B, A_c, A_f	0.825, 0.808, 0.923	0.825, 0.808, 0.923
Capital contribution $\Upsilon_c(q)A_c$ (initial $\to$ mask end, gha)	0.08364 $\to$ 0.04038	0.08364 $\to$ 0.04038
Fast-book contribution Y_f (initial $\to$ mask end, gha)	0.51000 $\to$ 0.78462	0.51000 $\to$ 0.78462
P_0 floor threshold ($A_{c0}=1.0$ ha, $\tau_g=5$ yr, $\Delta t=0.1$ yr)	2.955	2.955
P_0 floor threshold ($A_{c0}=1.0$ ha, $\tau_g=20$ yr)	2.955	2.954
P_0 floor threshold ($A_{c0}=1.0$ ha, $\tau_g=40$ yr)	2.955	2.954

Recompute-then-assert. The figure script (`make_fig_model.py`) regenerates the demonstration run and asserts $B(0)=0.594$ gha; at the end of the simultaneous $\dot{B}>0$, $\dot{A}_c<0$ window, $B=0.825$ gha, $A_c: 1.500 \to 0.808$ ha, $A_f: 0.600 \to 0.923$ ha, and the capital contribution $\Upsilon_c(q)A_c: 0.08364 \to 0.04038$ gha (a 51.7% decline); the fast-book contribution $Y_f = B - \Upsilon_c(q)A_c$ rises from 0.51000 to 0.78462 gha. It also checks a window of 150 ± 2 yr and an area-conservation residual $< 10^{-12}$ ha; it recomputes dimensionless k, the slope $b_{f0}k - b_c$ in gha ha^{-1}, $E_{\text{ceil}}=1.1384$ gha and the five balance points in ha, and asserts each against the stated value before the figure is written. The drawdown sweep (`drawdown_depth.py`) writes `si_drawdown_depth.csv`, from which Fig. 1c and Tables S4–S5 are produced. Eigenvalues and ceilings in Table 2a of the paper are computed from the closed-form J of Section S2.2 and agree with the numerical Jacobian of `work/verify_tab_conv_basin.py` to four decimals.

Well-posedness. The state space $\Omega = \{(A_f, A_c, A_r, q, P, D) : A_i \geq 0,\ q \in [0, q_{\max}],\ A_f + A_c + A_r = A_{tot},\ K \geq K_{\min}\}$ is forward invariant under the clamped scheme; the only non-smoothness is the exact positive part at $S=0$ and the minimum in Eq. (5), both of which are Lipschitz, so solutions exist and are unique for each history. Stability statements at $S=0$ are one-sided (deficit side).

S4 Extended model results

S4.1 Fast-land yield sweep and parameter sensitivity of the composition diagnostic

Table S3 extends Table 2a of the paper with the slope of the balance branch. The sign of Δb_{conv} is robust over a threefold range of the standing value: at $b_f = 0.85$ gha ha^{-1}, $\Upsilon_c(q^*) = 0.058, 0.066, 0.074$ gha ha^{-1} for $b_{G,c} = 0.4, 0.8, 1.2$ gha yr ha^{-1}, so $\Delta b_{\text{conv}} = +0.792, +0.784, +0.776$ gha ha^{-1}.

Table S3: Fast-land yield sweep at the baseline (all other parameters fixed): composition diagnostic at q^*, eigenvalues of the deficit-side matrix, slope of the balance branch and demand ceiling.

b_f (gha ha^{-1})	$\Delta b_{\text{conv}}(q^*)$ (gha ha^{-1})	λ_1 (yr^{-1})	λ_2 (yr^{-1})	$b_f k - b_c$ (gha ha^{-1})	E_{ceil} (gha)
0.60	+0.534	−0.0522	−0.975	0.277	0.804
0.85	+0.784	−0.0537	−0.998	0.414	1.138
1.20	+1.134	−0.0547	−1.014	0.605	1.606
2.00	+1.934	−0.0556	−1.029	1.041	2.675

S4.2 Drawdown-depth ladder

Table S4 gives the minimum capital land reached during endogenous-demand runs ($A_{f0} = 0.6$, $q_0 = 0.9$, $\tau_g = 20$, $\tau_p = 25\,\mathrm{yr}$, $\chi_r = 0.10$, $\Delta t = 0.1\,\mathrm{yr}$, $T = 400\,\mathrm{yr}$) on a grid of initial population P_0; the full grid (step 0.1) is in `si_drawdown_depth.csv`. The depth is continuous and monotone in P_0 and in A_{c0}. The initial population P_0 (cap) at which the minimum comes within 10^{-2}, 10^{-3} and 10^{-4} ha of the floor is 1.8, 1.9, 2.1 ($A_{c0} = 0.40\,\mathrm{ha}$); 2.2, 2.3, 2.4 ($A_{c0} = 0.70\,\mathrm{ha}$); 2.6, 2.7, 2.8 ($A_{c0} = 1.00\,\mathrm{ha}$); 2.8, 2.9, 3.0 ($A_{c0} = 1.18\,\mathrm{ha}$); the bisection with a 10^{-6} ha floor-proximity tolerance ($\Delta t = 0.05\,\mathrm{yr}$, $T = 1000\,\mathrm{yr}$) gives 2.41, 2.70, 2.98, 3.14 cap. The thresholds are identical to three decimals for $\chi_r = 0.10$ and $\chi_r = 0$ (2.387 and 2.955 cap at $A_{c0} = 0.40$ and 1.00 ha, $\Delta t = 0.1\,\mathrm{yr}$), and for $\tau_g = 5$, 20, 40 yr (Table S2). The minimum is reached at the moment the shortfall closes ($B \geq E$ for the first time), which occurs between 3 and 25 yr after the start depending on P_0.

Table S4: Minimum capital land $\min_t A_c$ (ha) reached during the run, by initial capital A_{c0} (rows, ha) and initial population P_0 (columns, cap); typed floor $A_c^{\min} = 0.05\,\mathrm{ha}$.

$A_{c0} \setminus P_0$	1.0	1.5	2.0	2.5	2.8	3.0	3.5	4.0
0.4	0.396	0.147	0.050	0.050	0.050	0.050	0.050	0.050
0.7	0.700	0.442	0.121	0.050	0.050	0.050	0.050	0.050
1.0	1.000	0.731	0.398	0.075	0.050	0.050	0.050	0.050
1.18	1.180	0.898	0.557	0.230	0.055	0.050	0.050	0.050

S4.3 Recovery metric and the role of the restoration flow

The recovery metric is $\mathcal{R}_c(T) = (A_c(T) - A_c^{\mathrm{deg}})/(A_c^{\mathrm{init}} - A_c^{\mathrm{deg}})$, with $\mathcal{R}_c = 0$ no recovery, 1 full recovery to the reference, < 0 continued decline and > 1 over-restoration. Table S5 evaluates it on the drawdown runs with $A_{c0} = 1.0\,\mathrm{ha}$ ($A_c^{\mathrm{init}} = 1.0\,\mathrm{ha}$, $A_c^{\mathrm{deg}} = \min_t A_c$ in ha, $T = 400\,\mathrm{yr}$): without restoration the capital book stays at its minimum (no return flow exists), with the surplus-gated flow it recovers partially, and at the mildest shortfall it over-restores ($\mathcal{R}_c = 1.28$), moving up the equilibrium family to a higher-capital, lower-biocapacity member while the supported population re-equilibrates downward.

Table S5: Recovery of the capital book after a demand-driven drawdown ($A_{c0} = 1.0\,\mathrm{ha}$, $A_{f0} = 0.6\,\mathrm{ha}$, $q_0 = 0.9$, $T = 400\,\mathrm{yr}$): initial population P_0 is in cap, initial footprint E_0 in gha, capital-land values in ha, and the recovery metric is dimensionless.

P_0	E_0	$\min_t A_c$	t(min) (yr)	$A_c(T)$, $\chi_r = 0$	$A_c(T)$, $\chi_r = 0.10$	$\mathcal{R}_c(T)$, $\chi_r = 0.10$
1.5	0.825	0.731	3.0	0.731	1.076	1.28
2.0	1.100	0.398	3.8	0.398	0.828	0.71
2.5	1.375	0.075	4.0	0.075	0.628	0.60
3.0	1.650	0.050	13.8	0.050	0.540	0.52
3.5	1.925	0.050	20.6	0.050	0.504	0.48
4.0	2.200	0.050	25.0	0.050	0.491	0.46

S4.4 Gate-sign phase dependence

The gate-sign comparison is evaluated from a non-equilibrium post-crash state ($A_c = 0.058\,\mathrm{ha}$ near the floor, $A_f = 0.6\,\mathrm{ha}$, dimensionless $q = 0.9$, $P = 1.2\,\mathrm{cap}$) rather than from the degraded equilibrium itself, because at the degraded $R_B = 1$ equilibrium $B = E$ and neither $[B - E]_+$ nor $[E - B]_+$ can fire. The reference is the healthy member of the equilibrium family reached from $A_{c0} = 0.9\,\mathrm{ha}$ ($A_c^{\mathrm{init}} = 0.8225\,\mathrm{ha}$, $P^{\mathrm{init}} = 1.489\,\mathrm{cap}$) and the degraded level is $A_c^{\mathrm{deg}} = 0.0513\,\mathrm{ha}$.

All rows use the base model ($T_b = 0$). Table S6 reproduces the record (package-local validation script `work/recovery_metric_and_tau_p_scan.py`).

Table S6: Restoration policy and gate sign: recovery metric at $T = 100$ and 900 yr, end-state books, population relative to the reference, extremes over the run and peak debt D (gha yr). Annual account values $B = E$ are in gha; land-book areas are in ha; population ratios and recovery metrics are dimensionless. Every row re-equilibrates to $R_B = 1$ ($E = B$); conservation holds to machine precision in every row.

Policy / gate	$\mathcal{R}_c(100)$	$\mathcal{R}_c(900)$	A_c	A_f	A_r	$B = E$	P/P^{init}	$\min A_r$	$\max A_c$	D_{peak}
none ($R_{rc} = 0$)	−0.00	−0.00	0.051	1.336	1.113	1.138	1.39	1.113	0.057	0.152
surplus-gated, $\chi_r = 0.02$	0.69	0.73	0.612	1.030	0.858	0.906	1.11	0.858	0.612	0.152
surplus-gated, $\chi_r = 0.05$	1.08	1.07	0.875	0.886	0.738	0.797	0.97	0.724	0.885	0.152
surplus-gated, $\chi_r = 0.10$	1.28	1.27	1.030	0.802	0.668	0.733	0.90	0.602	1.092	0.152
surplus, $\chi_r = 0.10$, target cap $A_c^* = 0.8225$	1.00	1.00	0.822	0.915	0.763	0.819	1.00	0.664	0.822	0.152
surplus, $\chi_r = 0.10$, conversion frozen	1.35	1.36	1.098	0.765	0.637	0.705	0.86	0.602	1.098	0.546
deficit-gated, $\chi_r = 0.05$	0.00	0.00	0.055	1.334	1.112	1.136	1.39	1.112	0.060	0.145
deficit-gated, $\chi_r = 0.10$	0.01	0.01	0.059	1.331	1.110	1.135	1.39	1.110	0.065	0.139
deficit, $\chi_r = 0.10$, conversion frozen	0.05	0.05	0.088	1.315	1.096	1.123	1.37	1.096	0.088	0.159

Reading. No restoration gives no recovery; the degraded state is a valid $R_B = 1$ equilibrium with a larger-than-reference population supported by fast land. Surplus-gated restoration recovers the capital book but at $\chi_r \geq 0.05$ restores it beyond the reference because the flow keeps running while $B > E$; the supported population falls modestly ($P/P^{\text{init}} = 0.97$–0.90). A target cap converts this into an exact return ($\mathcal{R}_c = 1.00$, $P/P^{\text{init}} = 1.00$). Deficit-gated restoration is the one gate that acts during the shortfall but is swamped by concurrent conversion at these rates ($\mathcal{R}_c \approx 0$); it is necessary but insufficient unless sized up or paired with a conversion cap. Freezing conversion under surplus-gated restoration raises the peak debt ($0.152 \to 0.546$) because the fast-land release channel can no longer close the shortfall while restoration over-supplies the lower-yield book; A_f falls under the freeze because restoration drains the reserve that maintains it. The earlier v43 wave-on recovery outcome that drove A_c to about 2.44 ha, reduced A_r to 0.027 ha (below its 0.10 ha floor), and drove returned B to 0.150 gha was not a valid model result: that solver applied T_b to the within-step yield but omitted it from returned B and the delayed B history. As a separate solver-regression check, not a reported scenario, the v44 bookkeeping correction applied to the same wave-on recovery initial state ($\Delta t = 0.05$ yr, $T = 900$ yr) gives $A_c^{\max} = 1.731$ ha, $A_r^{\min} = 0.321$ ha, and final $B = 0.899$ gha; the reserve floor is respected. The gate-sign table above uses the base model with $T_b = 0$. The separate controlled wave/no-wave debt comparison is reported in SI §S8.

S4.5 Regime table

Table S7: The composition diagnostic Δb_{conv} (gha ha^{-1}) as the decision condition for mask visibility. There is no interior area-based capacity maximum in the quality-based model; the only shape distinction is the sign of Δb_{conv}.

Regime	Condition	Behavior
composition-premium	$\Delta b_{\text{conv}} > 0$	the conversion contribution to accounting B is positive; net B also depends on other flows (the mask is possible in the baseline)
composition-neutral	$\Delta b_{\text{conv}} = 0$	first-order neutrality; non-generic
composition-discount	$\Delta b_{\text{conv}} < 0$	conversion lowers B at once; no mask; the liquidation sign is clear

S5 Empirical details

S5.1 Data and vintages

The annual NFA account values B and E are in gha; ratios and logarithmic changes are dimensionless. The aggregate growth of +23.0% ($\Delta \ln B = +0.207$) is anchored to the NFA aggregate world series (2025 edition). The land-type matrix (cropland, grazing land, forest products, fishing grounds, built-up land) incorporates a retroactive re-basing and gives +24.5% ($\Delta \ln B = +0.219$); the 1.5-percentage-point difference is an edition move, not a change in the underlying land facts. The attribution is robust across both: cropland accounts for $\approx 103\%$ of net growth against the aggregate series and $\approx 97\%$ against the land-type total, the residual band reflecting the edition move; the non-cropland book is flat as an endpoint change (1.008×; intra-period index up to 1.024×) in both. Population is from the same accounts (cap). The series are account-based (measured yields and conversion factors) and do not capture cumulative degradation; estimated biocapacity likely overstates the real base.

S5.2 Divisia decomposition, ϕ_0-robustness and the level fallback

With annual account values $X_t = \hat{b}_{f,t} A_{f,t}$ and $C_t = B_t - X_t$ in gha, the log-mean identity $\Delta \ln B = w_X(\Delta \ln b_f + \Delta \ln A_f) + w_C \Delta \ln C$ with $w_X = L(X_T, X_0)/L(B_T, B_0)$, $w_C = L(C_T, C_0)/L(B_T, B_0)$ holds exactly whenever X and C are positive at both endpoints. The dimensionless base-year calibration $\phi_0 = b_f(1961) A_f(1961)/B(1961)$ (the cropland share of 1961 biocapacity) is unidentified; Table S8 recomputes the decomposition over $\phi_0 \in [0.10, 0.34]$. The identity sums to +0.207 exactly in every row and the yield channel is the dominant positive term in every row, but the split is not pinned: w_X ranges 0.183–0.622, the 2022 fast-book share ranges ≈ 0.30–1.00, and the raw residual log-change ranges ≈ -0.04 to ≈ -6.8. Only the sign ordering is carried forward.

Table S8: Sensitivity of the proxy decomposition to the dimensionless base-year cropland share ϕ_0 (world, 1961–2022). Log-mean weights, log changes and weighted contributions are dimensionless.

ϕ_0	w_X	w_C	yield	area	residual	sum	dominant
0.10	0.183	0.795	+0.207	+0.029	−0.029	+0.207	yield
0.15	0.275	0.689	+0.310	+0.044	−0.148	+0.207	yield
0.19	0.349	0.600	+0.393	+0.056	−0.242	+0.207	yield
0.25	0.459	0.457	+0.517	+0.073	−0.384	+0.207	yield
0.30	0.550	0.315	+0.621	+0.088	−0.502	+0.207	yield
0.34	0.622	0.087	+0.702	+0.100	−0.594	+0.207	yield

The log-mean weighting fails once $C_{2022} \le 0$, which happens for

$$\phi_0 \;\ge\; \frac{B_{2022}}{B_{1961}\, e^{\Delta \ln A_f + \Delta \ln b_f}} \;\approx\; 0.339. \tag{S4}$$

In that regime the exact decomposition is the level one, $\Delta B = \Delta X + \Delta C$ with $\Delta X = \phi_0 B_{1961}(e^{\Delta \ln A_f + \Delta \ln b_f} - 1)$ and $\Delta C = \Delta B - \Delta X$, which holds for every ϕ_0, reproduces $\Delta B = +2.242 \times 10^9$ gha, and preserves the sign statement ($\Delta X > 0$, $\Delta C < 0$). Average-share weights ($\bar{s}_X = 0.375$, $\bar{s}_C = 0.625$) give +0.231, an $\approx 11\%$ overshoot, and are not used. The residual C is an accounting residual (grazing, forest products, fishing, built-up land, equivalence-factor and calibration effects), not a direct measurement of the model's effective area A_c or contribution $\Upsilon_c(q) A_c$; the instantaneous fast-book share s_X rises from 0.19 to 0.56 because the crop-yield index is applied to the whole fast book.

S5.3 Cropland-book calibration

The base year can be pinned directly from the measured NFA cropland biocapacity: $b_f(1961) = B_{\text{cropland,NFA}}(1961)/A_f(1961) \approx 0.93\,\text{gha}\,\text{ha}^{-1}$, corresponding to a cropland share of B of ≈ 0.128

in 1961, inside the robustness band of Table S8. Over 1961–2022 cropland biocapacity grew 2.85×, cropland area 1.17× and cropland biocapacity per hectare 2.43×; the model's representative $b_{f0} = 0.85\,\mathrm{gha\,ha^{-1}}$ is of the same order as the measured 1961 value and is not a fit.

S5.4 Observation-operator hierarchy

- *Fast provisioning land.* $A_f(t)$ = FAOSTAT arable plus permanent crops, element Area, world, 1961–2022 (ha) (FAO, 2023).
- *Fast-book yield.* $\tilde{b}_f(t)$ = FAOSTAT crop production per hectare of A_f, normalized to the base year and calibrated there against NFA cropland biocapacity, $b_f(t) = b_f(t_0)\tilde{b}_f(t)/\tilde{b}_f(t_0)$. FAOSTAT yields are physical output per hectare, not $\mathrm{gha\,ha^{-1}}$, and must be normalized or calibrated before any absolute statement; a cleaner proxy is aggregate crop production per hectare on a calorie or dry-matter basis.
- *Ecological-capital land.* A forest-area proxy, $A_c^{\mathrm{proxy}}(t)$, can be built from FAO/FRA forest area (ha) (FAO, 2020), with natural grassland, shrub and wetland added where available; before 1990, LUH2 (Hurtt et al., 2011) or HYDE 3.2 (Klein Goldewijk et al., 2017) can be scaled to FRA. This is a sentinel proxy, not a direct measure of the model's effective capital book (it omits soil carbon, wetland, fishery and non-forest regenerative capital). Where physical capital components are not separately measured, one can compute the accounting residual $C = B_{\mathrm{NFA}} - b_f A_f$; it must not be equated with A_c without an independent mapping and valuation.
- *Capital yield and regeneration.* $b_{G,c}G_c(q)$ is not observed in the NFA; physical proxies could include FRA growing stock or net annual increment, GAEZ potential productivity, or EarthStat/MapSPAM (Monfreda et al., 2008). The NFA aggregate does not separately observe $b_c A_c$ and $b_{G,c}G_c(q)A_c$. Even when b_f and A_f are independently estimated, the residual constrains only their sum, $\Upsilon_c(q)A_c$; separating capital area, quality, and regeneration requires an explicit observation operator and independent physical measurements.
- *Hierarchy.* (1) NFA land-type biocapacity with physical area and yield factors (best); (2) FAOSTAT area and yields as independent external proxies (used here); (3) remote-sensing land cover (ESA, 2017; Copernicus, 2019) or the reconstructions above with EarthStat/MapSPAM/GAEZ, for validation or extension.

  Table S9 summarizes the model-to-NFA mapping and its identification limits.

S5.5 Regeneration anchors

The capital-land regeneration timescale is a lumped recruitment-response timescale; the anchors bound a band and do not calibrate ρ_c. A lagged saturating recovery model fitted to plot-level age–above-ground-biomass records of Neotropical secondary forests (2ndFOR; Poorter et al., 2016, 1,334 plots, 41 chronosequences, 35 good fits) gives a median $t_{50} \approx 30\,\mathrm{yr}$ and a recovery timescale $\tau_r \approx 38\,\mathrm{yr}$; the same model fitted to deplete-then-rebuild trajectories of collapsed fish stocks (RAM Legacy B/B_{MSY}, 33 episodes) gives $t_{50} \approx 5.6\,\mathrm{yr}$ and $\tau_r \approx 3.4\,\mathrm{yr}$ (cf. Hutchings and Reynolds, 2004; Neubauer et al., 2013); soil organic carbon re-equilibrates after ≈ 17–23 yr following land-use change (Poeplau et al., 2011). Forest biomass recovers to about half of old-growth values by ≈ 20 yr and to 90% at a median ≈ 66 yr (Poorter et al., 2016) (the 66–95 yr tail lies beyond the core band); only ≈ 29% of collapsed fish stocks recovered to 50% within 5–15 yr (Hutchings and Reynolds, 2004). The two fitted legs bracket the effective timescale (forest slow, fishery fast) with soil between them; the band is ≈ 25–33 yr (representative ≈ 29 yr; order 10–40 yr). These are statements about the regeneration rate, not the delay τ_g. In the one-stock comparator the forest value lies at or above that model's ≈ 18–20 yr cliff; in the two-land model

Table S9: Conceptual mapping from two-land model quantities to NFA information. The rows describe a proxy/observation operator, not an identity between an NFA category and a model state. Annual account values are in gha; land areas are in ha.

Model quantity	Closest NFA/account information	Interpretation and identification limit
Fast area and account A_f, $Y_f = b_f A_f$	Cropland biocapacity; FAOSTAT arable and permanent-crop area and yield proxies used here	Cropland biocapacity anchors the fast account at the base year. FAOSTAT yields are physical output per hectare and require normalization/calibration; the mapping is not a one-to-one identity.
Capital area, quality and account A_c, q, $Y_c = \Upsilon_c(q)A_c$	No separate NFA category; the residual $C = B - Y_f$ includes non-cropland account values	In the model $B - Y_f = Y_c = \Upsilon_c(q)A_c$; empirically, the residual also reflects built-up land, equivalence-factor, vintage and calibration effects. With an independently observed fast book it constrains the composite contribution, not A_c or q separately, and it is not a physical-stock measure.
Reserve area A_r	No direct NFA account counterpart	A_r closes the model's conserved-area ledger; it is not recoverable from the NFA aggregate.

Table S10: Observability ladder: what is measured, what is an identifiable combination, and what is not identifiable.

Tier	Quantities	Status
Observable physical area / land cover	A_f, A_{tot}, and mapped land-cover classes [ha]; forest area as A_c^{proxy}	physical source measures are direct; mapping to effective A_c is not
Observable (population)	P [cap]	direct
Observable (physical productivity proxy)	$\tilde{b}_f(t)$ [product $\text{ha}^{-1}\,\text{yr}^{-1}$]	measured; normalized to obtain b_f
Observable (annual account value, under a convention)	B, E [gha]	only under the NFA accounting convention; gha is value-weighted
Aggregate-derived combination	$C = B - b_f A_f = \Upsilon_c(q)A_c$ [gha]	identifiable only when B, b_f, and A_f are independently available; it is not a physical-stock measure
Effective capital value	$\Upsilon_c(q) = C/A_c$ [gha ha^{-1}]	requires an independent observation/mapping of A_c; not identified by the NFA total alone
Not separately identifiable	b_c, $b_{G,c}$, ρ_c, and q individually	with independent A_c and q observations over a useful range, at most b_c and $b_{G,c}\rho_c$ can be separated; $b_{G,c}$ and ρ_c remain confounded
Debt channel	α and D	requires an independent debt or yield-loss proxy

the drawdown boundary is a demand threshold and the anchors bound a band rather than locate a cliff.

S5.6 Recursive identifiability in the land-type accounts

In the land-type vintage the non-cropland total is flat at the endpoint (1.008×) but is the net of declining forest-products (0.90×), grazing (0.98×) and fishing (0.96×) biocapacity – together $\approx -6.4 \times 10^8$ gha – and an 8.63× rise in built-up land; contributions to the +24.5-point aggregate growth are cropland +23.8, built-up +7.3, forest products, grazing and fishing −6.6 (gross positive +31.1). These categories are accounting flows and land classes, not direct measurements of the model's effective area A_c or the contribution $\Upsilon_c(q)A_c$; they do not demonstrate a measured decline in A_c, but they show that an aggregate conceals composition at every level at which it is used.

S5.7 Physical-stock scoreboard and interpretation

A component-level physical inventory should be maintained as a vector of separately defined accounts, with stock or extent distinguished from annual flow and pressure. Table S11 gives an illustrative template rather than a new dataset or a calibration of the model. Each row should carry its reference year, spatial boundary, physical unit, absolute and relative change, uncertainty, data vintage, and any ecological floor or reference range. The dimensions are interpreted jointly, not added: for example, forest area and biomass over the same mapped area are nested measures, while catch and spawning-stock biomass are a flow and a stock. A single composite should be constructed only after declaring its valuation basis, normalization, weights, and permitted substitution. This is consistent with physical ecosystem-extent and condition accounts being reported separately from ecosystem-service and monetary presentations in SEEA Ecosystem Accounting (United Nations et al., 2021).

Table S11: Illustrative component-level physical-stock scoreboard. Units and account boundaries must be specified for the chosen data source; stock, extent, condition, flow, and pressure columns are not additive.

Component	Physical extent, stock, or condition	Annual flow or pressure (separate)	Interpretation safeguard
Forests and woodlands	Extent (ha); growing stock or biomass (m^3 or tonnes C); stated condition indicators	Net annual increment, harvest, or deforestation ($m^3\,yr^{-1}$ or tonnes $C\,yr^{-1}$)	Area is not biomass or condition; forest-products biocapacity is not a direct stock measure.
Soils	Soil organic carbon stock (tonnes C) or density (tonnes $C\,ha^{-1}$) for a stated depth; structure/erosion condition	Carbon accumulation or loss and soil erosion (tonnes $C\,yr^{-1}$ where measured)	Hold depth, bulk density, land cover, and sampling method consistent across time.
Wetlands	Extent (ha) and an explicit ecological-condition class or index	Carbon accumulation, water storage change, or nutrient flux (source-specific units per year)	Declare overlap with forest and other land-cover classes; do not add overlapping extents.
Fisheries	Spawning-stock biomass by species or stock (tonnes); age or size structure where available	Recruitment, catch, and landings (tonnes yr^{-1})	Catch/landings are removals, not stock; retain stock-specific definitions and uncertainty.
Freshwater systems	Surface-water or groundwater storage (m^3), aquifer level, and quality indicators	Recharge and withdrawal ($m^3\,yr^{-1}$)	Separate basin and aquifer boundaries; do not substitute annual flow for stored volume.

In the two-land model, A_c is an effective area state (ha) and q a separate capacity-quality index; neither is directly observed in the NFA aggregate. Physical components can inform an

observation operator for these effective states only when their mapping and aggregation rule are declared. In particular, the NFA residual $C = B - b_f A_f$ must not be equated with a measured physical capital stock without independent area, condition, and valuation information.

S6 Policy extensions

Half-Earth and reservation. The Half-Earth proposal (Wilson, 2016) caps human-usable annual biocapacity account value (gha) at half of the total. In the two-land setting this is a reservation $E \leq \sigma B$ with dimensionless σ the human-available share, enforced either as a cap on the fast book or on the carrying capacity; with $\sigma_f, \sigma_c < 1$ in Eq. (4) the shortfall is computed against the reserved flows, so conversion begins earlier for a given demand and the ceiling of Proposition 2 is lowered to σ-weighted values. On the nominal path a reservation that keeps $E \leq \sigma B$ succeeds with no liquidation. This is a nominal-path consequence until it is certified against a declared disturbance class (persistent productivity, b_f or ρ_c shocks) under a preregistered retention rule (a governance module is retained only if it improves a declared protection-and-supply score), with the model's declared fit defects converted into an erosion margin on the closed loop's contraction rate; until then it is a property of the nominal calibration, not a robustness claim.

Typed floors and freeze conversion. Reserve $A_r \geq A_r^{\min}$, capital viability $A_c \geq A_c^{\min}$, arable ceiling $A_f \leq A_f^{\max}$ and $K \geq K_{\min}$ are separate constraints. Freezing conversion (capping A_f at $A_f^{\max}$ or flooring A_c at $A_c^{\min}$) is the single operator that guarantees $A_c \geq A_c^{\min}$; the conversion rate saturates as $A_c \to A_c^{\min}$ by construction of Eq. (5). A policy that observes only the aggregate B cannot guarantee capital-land integrity: the failure is informational (aggregation relative to the typed floor), not dynamical.

Institutional signature. In the mobilizing/protective taxonomy of the delay companion (Abaee, 2026a), freeze-conversion and surplus-gated restoration are protective; the mobilizing sign (demand responding to a perceived shortfall by converting more) is the destabilizing one; deficit-gated restoration is not unambiguously protective because it can deepen the short-run shortfall (Section S4.4).

Weak and strong sustainability reading. In the material-cycle reading of the ledger companion (Abaee, 2026c), weak sustainability is the idealized regime in which substitution and regeneration jointly redistribute matter so that it is used as it arises – substitution admissible only through an identified physical pathway, never merely because a weighted aggregate permits it – and strong sustainability is "the regime in which that closure fails – either because no identified physical pathway re-routes the extracted matter, or because depletion outruns the rate at which such a pathway could be deployed", so that a separately binding floor on a critical stock binds. The composite annual account value B (gha) is the weak-sustainability index: the linear, perfectly substitutable member of the aggregator family of the assessment companion (Lemma A of Abaee, 2026b); it can satisfy its aggregate floor while the typed floor on A_c is violated by exactly the amount that the fast-land yield gains against the capital loss, and in that family no positive-elasticity aggregator is uniformly safe (its Theorem S2(ii)). Under exact-tube semantics the intermediate passage during which A_c falls is itself a violation even if the endpoint recovers.

Reporting rule. A growth headline for total biocapacity carries no information about whether cropland growth is compensating for stagnation or offsetting changes in the other components. The informative assessment states, at every level at which an aggregate is used: component endpoint ratios, log contributions, attribution shares, data vintage and endpoint sensitivity.

S7 The one-stock comparator

Earlier versions of this work developed a single-stock model that the two-land model supersedes. It is retained here as a comparator so that quantities that circulate for it are not mistaken for two-land results. The comparator has three states (A, P, D) and one algebraic relation:

$$G(A) = \rho A(1 - A/A_{\max}), \qquad B = bA + b_G G(A), \qquad E = eP, \tag{S5}$$

$$\dot{A} = G\big(A(t-\tau_g)\big) - [E - \sigma bA]_+ / b_G, \tag{S6}$$

$$\dot{P} = rP\big(1 - P/K(t-\tau_p)\big), \qquad K = B/e, \tag{S7}$$

$$\dot{D} = [E - B]_+ - \eta D, \qquad b = (b_0 + T_b)e^{-\alpha D}, \tag{S8}$$

Here A and $A_{\max}$ are comparator-specific stock units and $G(A)$ has units stock unit yr^{-1}; consequently $b, b_0, T_b, \Delta b$ are gha per stock unit and b_G is gha yr per stock unit, so bA and $b_G G(A)$ are each gha. B and E are annual GFN/NFA account amounts in gha, P and K are in cap, and e is gha cap^{-1}. The shares σ are dimensionless. Time and the delays are in yr; ρ, r, η and the one-stock logistic-wave steepness κ are rates in yr^{-1} (κ is distinct from the two-land κ_w); D is a leaky time-integral in gha yr, so $\dot{D}$ and ηD are gha; and α is gha^{-1} yr^{-1}. The figures share a numerical ordinate only for illustration; A, P, B and D retain distinct units.

With demand met first from the flow bA, the shortfall liquidates the standing stock at rate $1/b_G$. Its properties, each of which was re-derived under the two-land structure and found not to carry over, are the following.

- *MSY and fold.* $B(A)$ has an interior maximum (the MSY of Schaefer, 1954) only in the capital-dominated regime $b_G\rho > b$; a fixed liability above $B_{\max}$ produces a saddle-node, the catastrophic-shift structure of Scheffer et al. (2001). At the comparator baseline ($b_G\rho = 0.04 < b = 0.5$ in gha per comparator-specific stock unit) the fold is absent. In the two-land model Y_c is linear in A_c and $B_{eq}(A_c)$ is monotone; there is no fold in any regime (Proposition 2).

- *The equilibrium family $P = B(A)/e$ and $\det J = 0\,\mathrm{yr}^{-3}$.* Under full harvest ($\sigma = 1$) the comparator has a neutral continuum of equilibria on which the Jacobian is exactly singular. The two-land model has a one-parameter family in the *land allocation* (indexed by A_c on the no-conversion face, with $A_r/A_f = \eta_f/\mu$), not a free P, and no identically singular Jacobian.

- *The $+0.6$ eigenvalue and the two-gain Hopf classification.* The comparator's deficit-region linearization has a positive real root ($a_{11} = G'(A^*) + b/b_G \approx 0.608$–$0.625\,\mathrm{yr}^{-1}$ at its baseline) and a dimensionless comparator-specific two-gain rule $\chi_{1s} = q/(\rho - 2q)$ (here q is the comparator's feedback-rate coefficient in yr^{-1}, not the two-land quality index) selecting which delay destabilizes (τ_g-only for $\chi_{1s} > 1$, τ_p-only for $\chi_{1s} < 1$, with critical delays $\tau_m \approx 85.4\,\mathrm{yr}$ and $\tau_p \approx 231\,\mathrm{yr}$). None of this is a two-land quantity: the two-land deficit-side spectrum is real and negative ($-0.054\,\mathrm{yr}^{-1}$, $-1.0\,\mathrm{yr}^{-1}$) and constant along the branch.

- *The regeneration cliff and the masking window.* The comparator shows a recover/collapse boundary at $\tau_g \approx 18$–$20\,\mathrm{yr}$ and a productivity-illusion window of $\approx 5.4\,\mathrm{yr}$ ($A : 1.00 \to 0.86$ comparator-specific stock units while $B : 0.578 \to 0.647\,\mathrm{gha}$; the mask vanishes for initial deficits $\gtrsim 0.075\,\mathrm{gha}$). In the two-land model the boundary is a demand threshold insensitive to τ_g (Table S2) and the composition window is set by λ_1 and Δb_{conv} (paper, Section 3.1). The dimensionless recovery fractions $0.54 \to 0.52 \to 0.50$ for $\tau_g = 5 \to 25 \to 40\,\mathrm{yr}$ quoted in earlier versions were computed with an area-based intermediate formulation and are superseded by the identical-to-10^{-3} result of Table S2.

The four selected comparator figures reproduced here are numbered S1–S4. The broader historical supplementary figure set (S1–S15) remains preserved in the repository at `agent 2 productivity illusion/supplementary/FIGURES/`; only figures used by the current SI are included in this package.

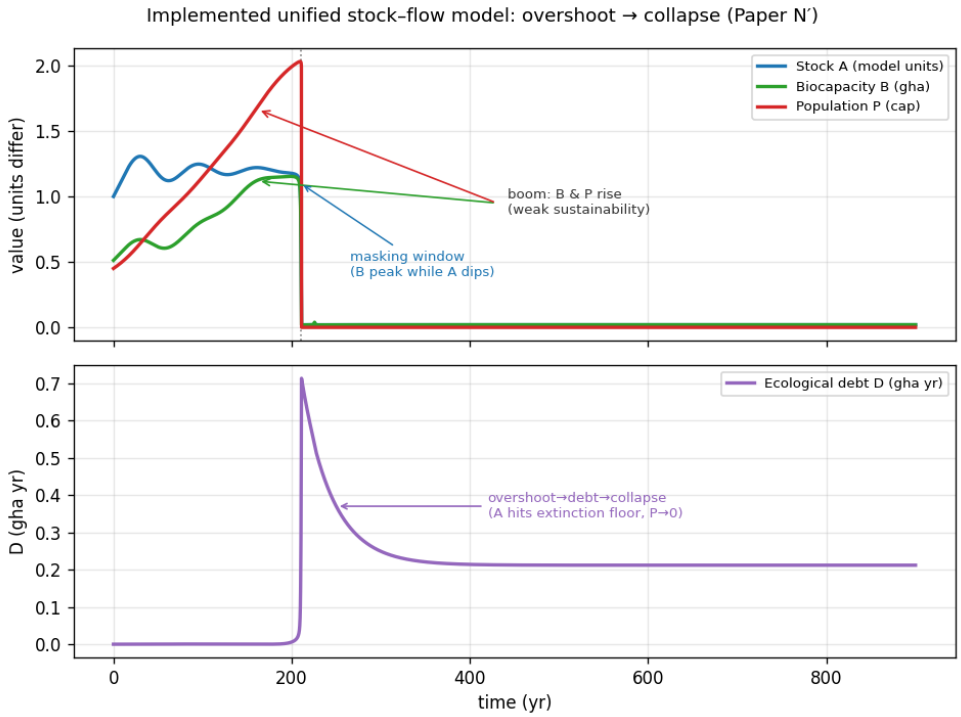


Figure S1: One-stock comparator: top panel shows stock A (model-specific stock units), population P (cap) and annual biocapacity B (gha); the lower panel shows leaky debt D (gha yr). The footprint $E = eP$ is a driver and is not plotted. The shared numerical ordinate is illustrative, not a common physical scale. Comparator only; not a two-land result.

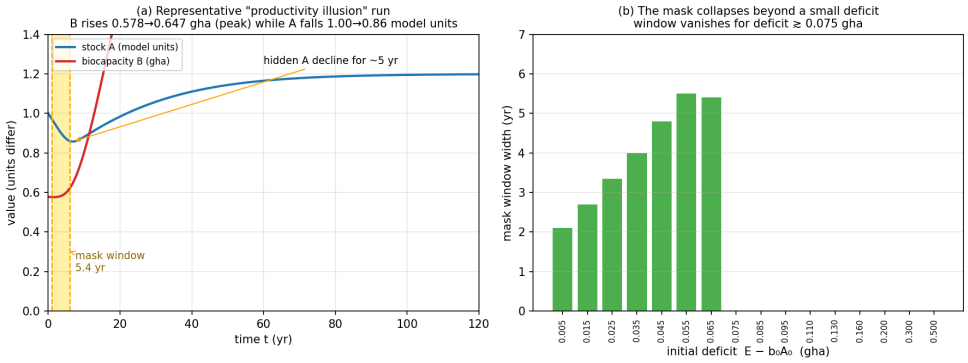


Figure S2: One-stock comparator: the productivity-illusion window (A falls from 1.00 to 0.86 model units while B rises from 0.578 to 0.647 gha; window ≈ 5.4 yr; the mask vanishes for deficits $\gtrsim 0.075$ gha). The two-land composition window of the paper (Fig. 1a) is a different object with a different mechanism.

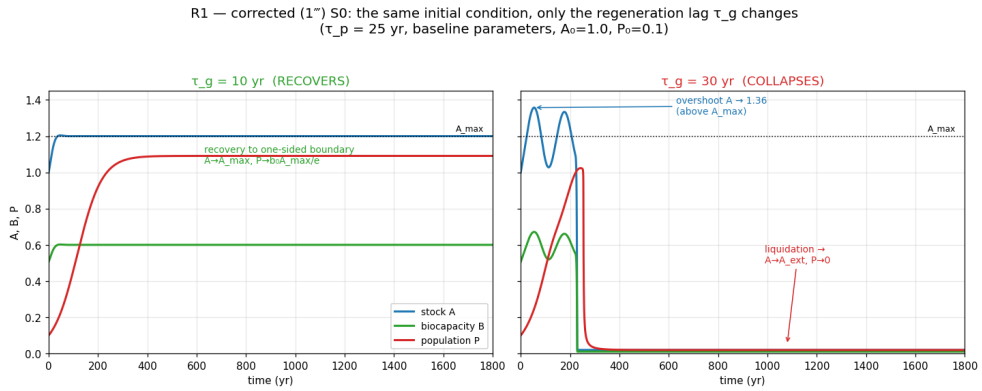


Figure S3: One-stock comparator: recovery versus collapse as a function of regeneration delay τ_g (yr; the $\approx$ 18–20 yr cliff). The plotted series use a shared numerical ordinate only for illustration: stock A is in comparator-specific units, biocapacity B in gha and population P in cap. Not present in the two-land model.

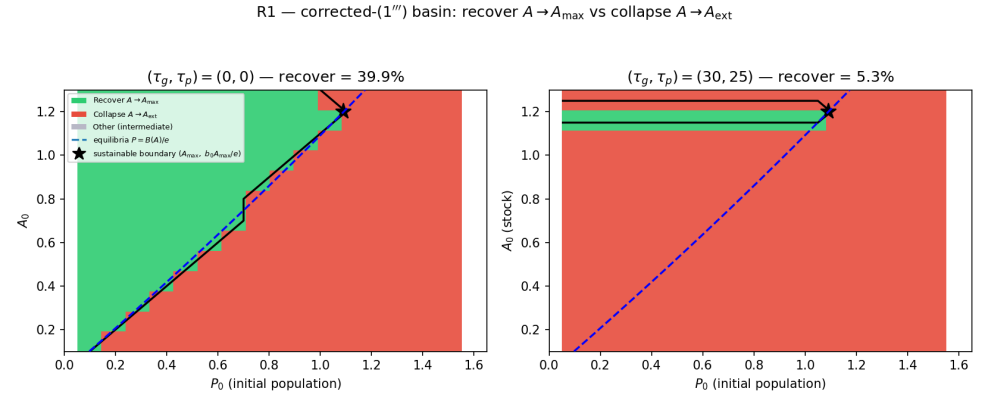


Figure S4: One-stock comparator: basin of recovery at the comparator baseline. The axes are initial population P_0 (cap) and initial stock A_0 (comparator-specific stock units); the equilibrium guide $P = B(A)/e$ is in cap, with annual account value B in gha and e in gha cap^{-1}. In the two-land model the corresponding object is the continuous drawdown-depth surface of Fig. 1c and Table S4.

S8 Controlled technology-wave experiment

This controlled extension tests whether an active T_b wave can produce a larger leaky-debt peak than an otherwise matched no-wave run. The pair uses endogenous demand $E = eP$ (so the surplus-gated R_{rc} is active only when $B > E$), $A_{tot} = 2.5\,\mathrm{ha}$, $A_{f0} = 0.6\,\mathrm{ha}$, $A_{c0} = 0.9\,\mathrm{ha}$, $q_0 = 0.9$, $D_0 = 0.0\,\mathrm{gha\,yr}$, and $P_0 = 1.0$ or $1.2\,\mathrm{cap}$. Shared parameters are $b_{f0} = 0.85\,\mathrm{gha\,ha^{-1}}$, $b_c = 0.05\,\mathrm{gha\,ha^{-1}}$, $b_{G,c} = 0.8\,\mathrm{gha\,yr\,ha^{-1}}$, $\rho_c = 0.08\,\mathrm{yr^{-1}}$, $\alpha = 0.03\,\mathrm{gha^{-1}\,yr^{-1}}$, $\eta = 0.05\,\mathrm{yr^{-1}}$, $\eta_f = 0.05\,\mathrm{yr^{-1}}$, $\mu = 0.06\,\mathrm{yr^{-1}}$, $e = 0.55\,\mathrm{gha\,cap^{-1}}$, $r = 0.02\,\mathrm{yr^{-1}}$, $\tau_g = 20\,\mathrm{yr}$, $\tau_p = 25\,\mathrm{yr}$, $\tau_{conv} = 1\,\mathrm{yr}$, and $\chi_r = 0.10\,\mathrm{gha^{-1}\,yr^{-1}}$. The wave parameters are $\Delta b = 1.0\,\mathrm{gha\,ha^{-1}}$, $\kappa_w = 0.1\,\mathrm{yr^{-1}}$, and $t_{wave} = 40\,\mathrm{yr}$; the horizon is $T = 250\,\mathrm{yr}$. Within each pair, the physical initial state, parameters, wave-free pre-run history, and solver are identical; only the T_b switch differs. The logistic is already nonzero at $t = 0$ ($T_b(0) = 0.017986\,\mathrm{gha\,ha^{-1}}$), so the derived initial account is $B(0) = 0.560184\,\mathrm{gha}$ without the wave and $0.570976\,\mathrm{gha}$ with it. This small initial difference is part of the declared forcing, not a change to A_f, A_c, q, P, or D; T_b reaches half its increment at the marked midpoint $t_{wave} = 40\,\mathrm{yr}$.

Figure S5 shows the $P_0 = 1.0$ pair. Its all-time debt maximum occurs after the wave midpoint in both runs: $D_{max} = 0.008126\,\mathrm{gha\,yr}$ at $86.15\,\mathrm{yr}$ without the wave, versus $0.020205\,\mathrm{gha\,yr}$ at $115.60\,\mathrm{yr}$ with it. At $P_0 = 1.2$, by contrast, the all-time maximum occurs early and is lower with the wave, while the maximum over $t \geq t_{wave}$ is higher. Table S12 keeps these all-time and post-midpoint definitions separate. We call $t \geq t_{wave}$ the post-midpoint window; it is not an assertion that a logistic wave has ended at t_{wave}. The full metrics and trajectories are deposited as `work/tb_wave_debt_metrics.csv`, `work/tb_wave_debt_timeseries_P0_1p0.csv`, and `work/tb_wave_debt_timeseries_P0_1p2.csv`; `work/tb_wave_debt_scan.py` regenerates them and the figure.

Table S12: Matched technology-wave/no-wave debt maxima. Entries are D_{max} in gha yr, with the time of the maximum in years in parentheses; the ratio is wave/no-wave. "All" is the maximum over $0 \leq t \leq 250\,\mathrm{yr}$; $t \geq 40$ is the post-midpoint maximum. The $P_0 = 1.0$ all-time peaks both occur after the midpoint.

P_0 (cap)	quality update	Δt (yr)	window	no wave D_{max} (time)	wave D_{max} (time)	ratio
1.0	delayed	0.050	all	0.008126 (86.15)	0.020205 (115.60)	2.487
1.0	delayed	0.025	all	0.008105 (86.20)	0.020195 (115.65)	2.492
1.0	current	0.050	all	0.008326 (85.70)	0.020168 (115.60)	2.422
1.2	delayed	0.050	all	0.061218 (1.65)	0.051037 (1.45)	0.834
1.2	delayed	0.050	$t \geq 40$	0.010854 (81.70)	0.025184 (112.55)	2.320
1.2	delayed	0.025	all	0.060732 (1.625)	0.050582 (1.475)	0.833
1.2	delayed	0.025	$t \geq 40$	0.010808 (81.775)	0.025165 (112.625)	2.328

The $P_0 = 1.0$ result is a declared sensitivity case, not a refit or a universal effect: wave/no-wave all-time peak ratios persist under step halving (2.487 at $\Delta t = 0.05\,\mathrm{yr}$; 2.492 at $0.025\,\mathrm{yr}$) and under the current-quality solver (2.422 at $0.05\,\mathrm{yr}$). At the reference $P_0 = 1.2$, the corresponding all-time ratios are 0.834 and 0.833, whereas the post-midpoint ratios are 2.320 and 2.328. All reported debt maxima change by less than 0.9% under step halving. Maximum area-conservation residual is $4.44 \times 10^{-16}\,\mathrm{ha}$; the minimum reserve in the $P_0 = 1.0$ wave run is $0.286\,\mathrm{ha}$, above the $0.10\,\mathrm{ha}$ floor. In this sensitivity pair A_c does not fall below its initial $0.9\,\mathrm{ha}$, so the experiment isolates a debt rebound and is not evidence of the capital drawdown shown in the main frozen-demand run. The regression test also confirms that an integer input $D_0 = 0$ still produces a floating-point debt state and finite debt accumulation. D is the model's leaky time-integral of shortfall, not an observed NFA or physical-stock variable. These conditional trajectories demonstrate a possible delayed rebound, while the $P_0 = 1.2$ contrast shows why the all-time peak cannot be substituted for the post-midpoint response.

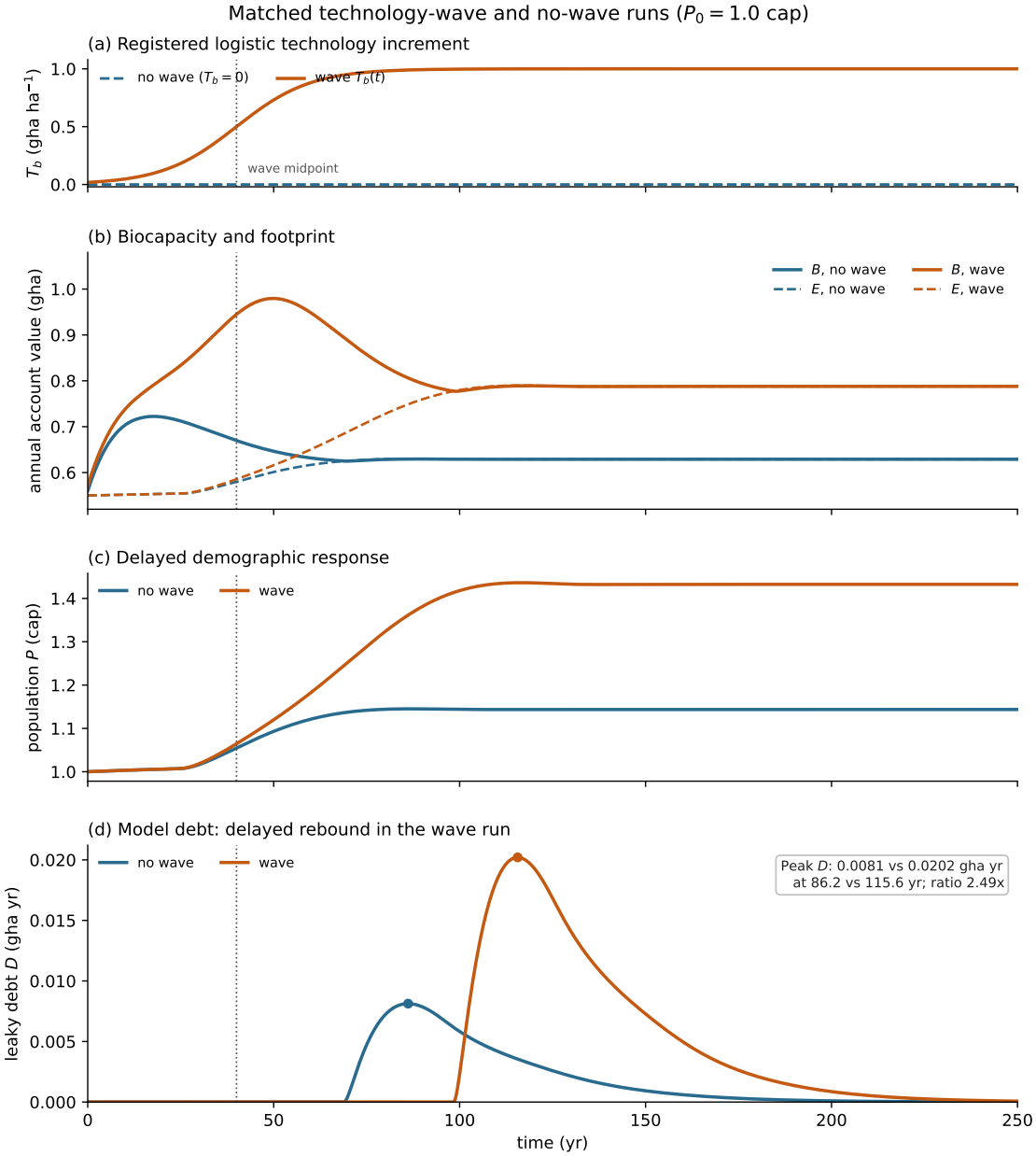


Figure S5: Matched endogenous-demand technology-wave/no-wave simulation at $P_0 = 1.0$ cap. Parameters and initial conditions are listed in SI §S8; only T_b is switched on or off. (a) The active logistic increment $T_b(t)$, with the dotted line at its midpoint $t_{wave} = 40$ yr; (b) annual account values B and E (gha); (c) population P (cap); (d) leaky model debt D (gha yr). Markers denote all-time maxima, which occur at 86.15 yr without the wave and 115.60 yr with it; peak debt is 0.008126 versus 0.020205 gha yr. The plotted wave begins with its small logistic tail at $t = 0$ and approaches $\Delta b = 1.0$ gha ha^{-1}; the dotted line marks the midpoint, not an end date.

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